LP3 MIDSEM MOCK PRACTICAL

Roll no: 41169

Assignment A3:

Write a program to solve a 0-1 Knapsack problem using dynamic programming or branch and bound strategy.

```
#include <bits/stdc++.h>
using namespace std;

struct Item { int w, v; };

int main() {
    int n, W;
    cout << "Enter number of items and capacity: ";
    cin >> n >> W;
    vector<Item> a(n);
    cout << "Enter weight and value of each item:\n";
    for (auto& it : a) cin >> it.w >> it.v;

    vector<vector<int>> dp(n + 1, vector<int>(W + 1, 0));
    for (int i = 1; i <= n; i++) {
        for (int c = 0; c <= W; c++) {
            dp[i][c] = dp[i - 1][c];
            if (a[i - 1].w <= c)
                dp[i][c] = max(dp[i][c], dp[i - 1][c - a[i - 1].w] + a[i - 1].v);
        }
    }
    cout << "Max value: " << dp[n][W] << "\nSelected items (weight, value):\n";

    for (int i = n, c = W; i > 0; i--)
        if (dp[i][c] != dp[i - 1][c]) {
            cout << "Item " << i << ": (" << a[i - 1].w << ", " << a[i - 1].v << ")\n";
            c -= a[i - 1].w;
        }
}
```

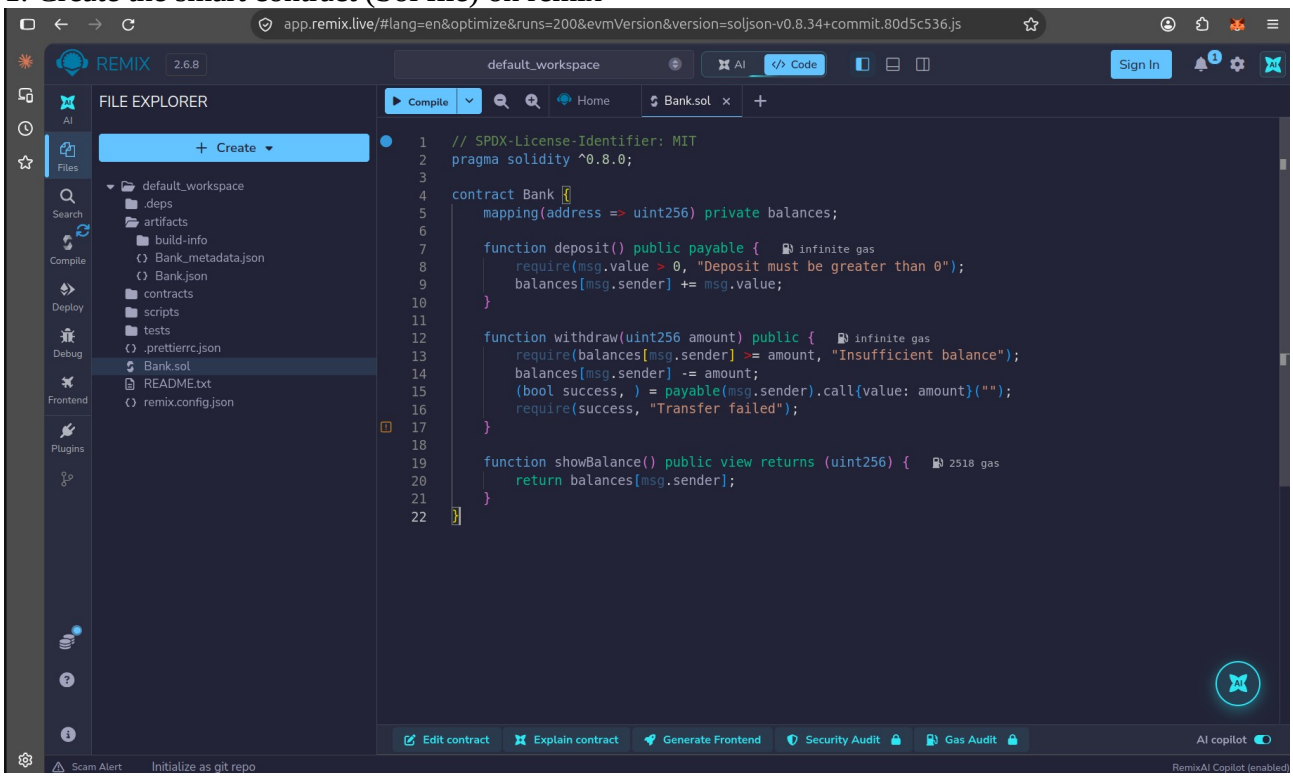
OUTPUT:

```
chinmayee@chinmayeeHyperXOMEN15:~/Chinmayee/College/LP3$
chinmayee@chinmayeeHyperXOMEN15:~/Chinmayee/College/LP3$
chinmayee@chinmayeeHyperXOMEN15:~/Chinmayee/College/LP3$ g++ LP3A3.cpp -o a
chinmayee@chinmayeeHyperXOMEN15:~/Chinmayee/College/LP3$ ./a
Enter number of items and capacity: 5
40
Enter weight and value of each item:
4
2
6
3
1
4
2
5
6
3
Max value: 17
Selected items (weight, value):
Item 5: (6, 3)
Item 4: (2, 5)
Item 3: (1, 4)
Item 2: (6, 3)
Item 1: (4, 2)
chinmayee@chinmayeeHyperXOMEN15:~/Chinmayee/College/LP3$
```

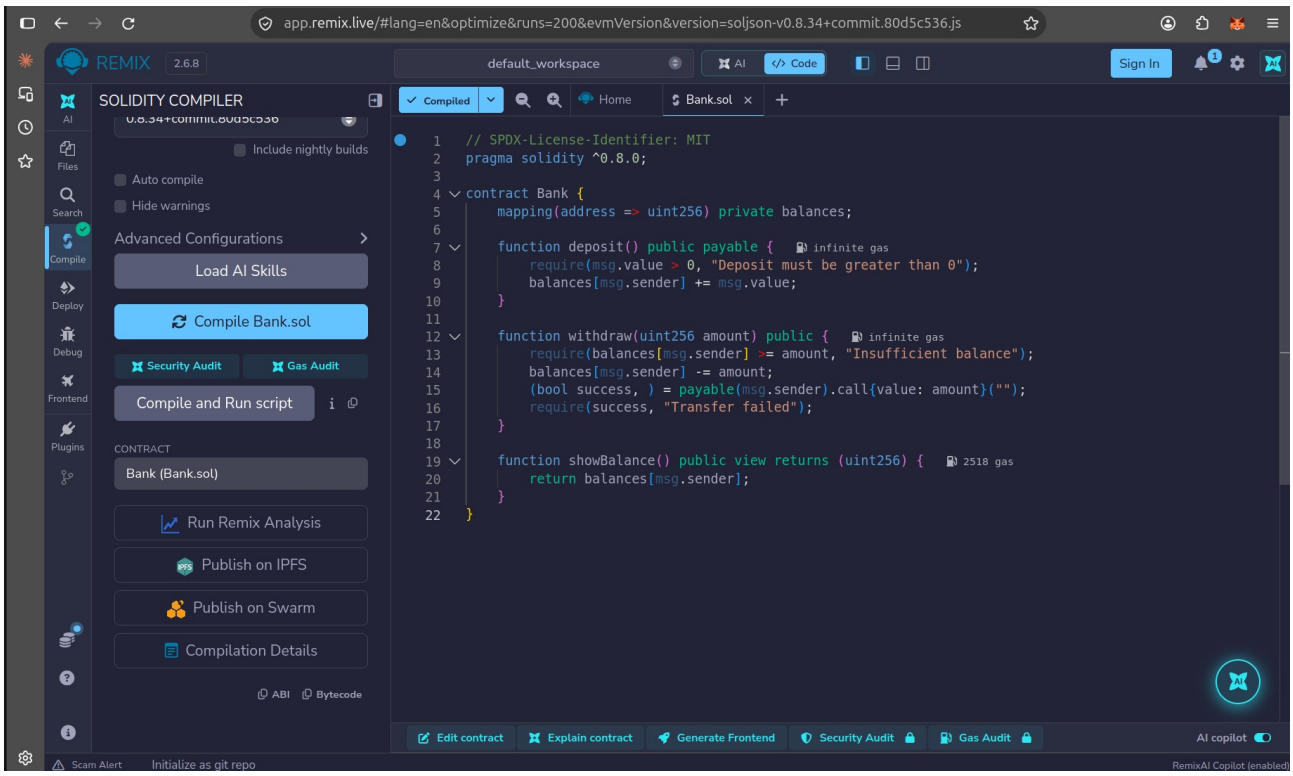
Assignment C2:

Create your own wallet using MetaMaskfor crypto transactions.

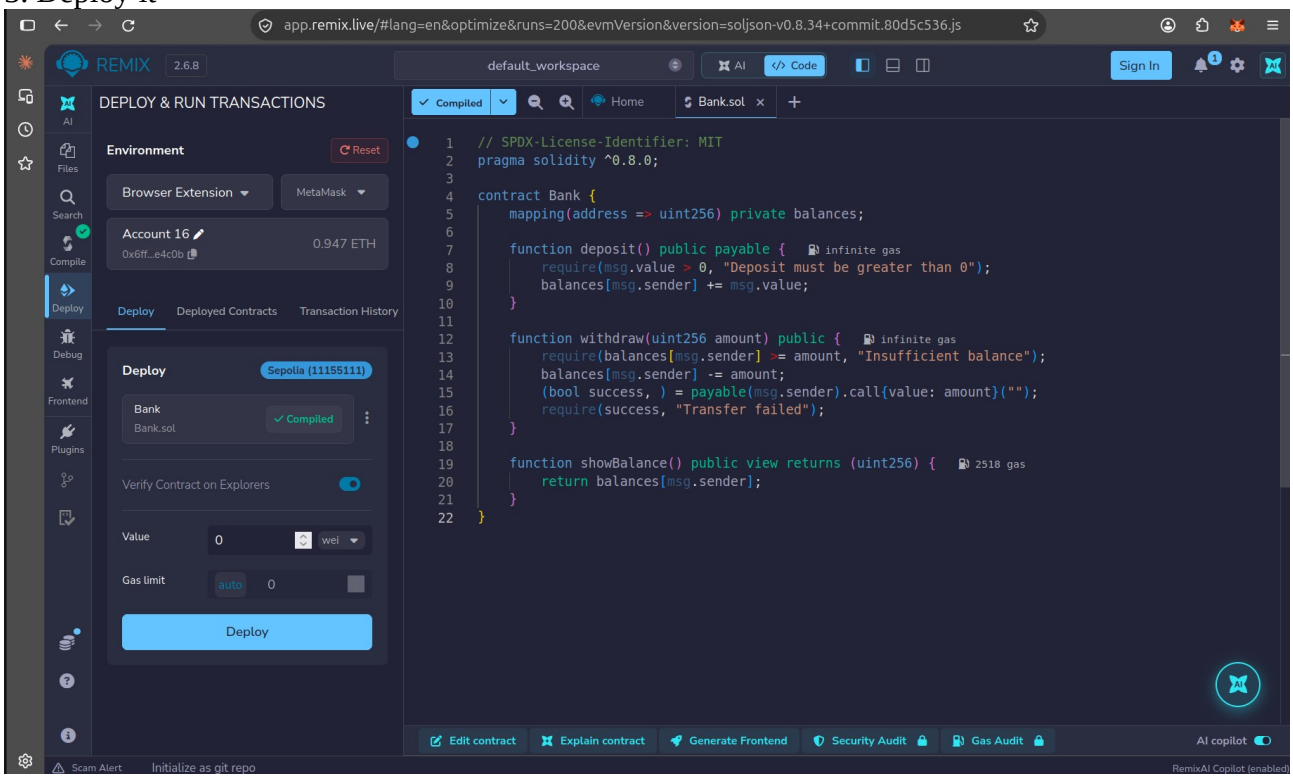
1. Create the smart contract (Sol file) on remix



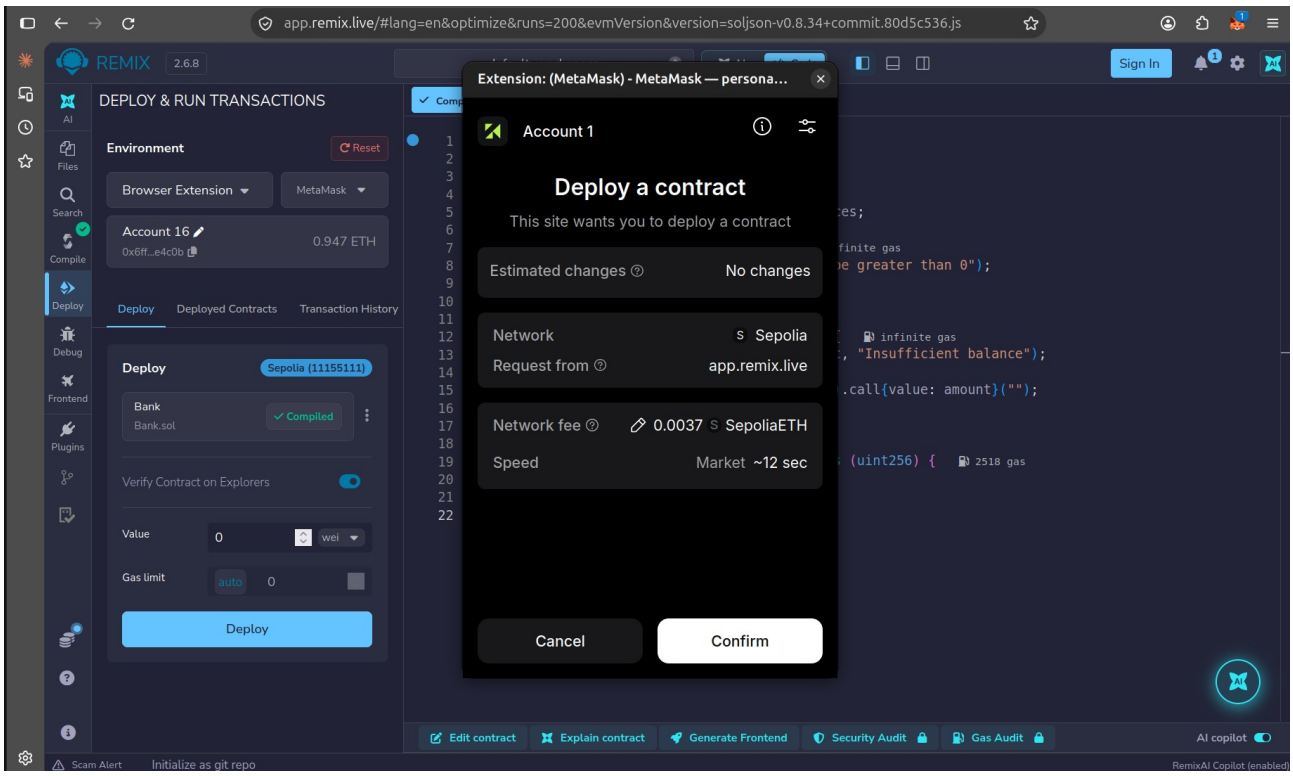
2. Compile the code on compile tab



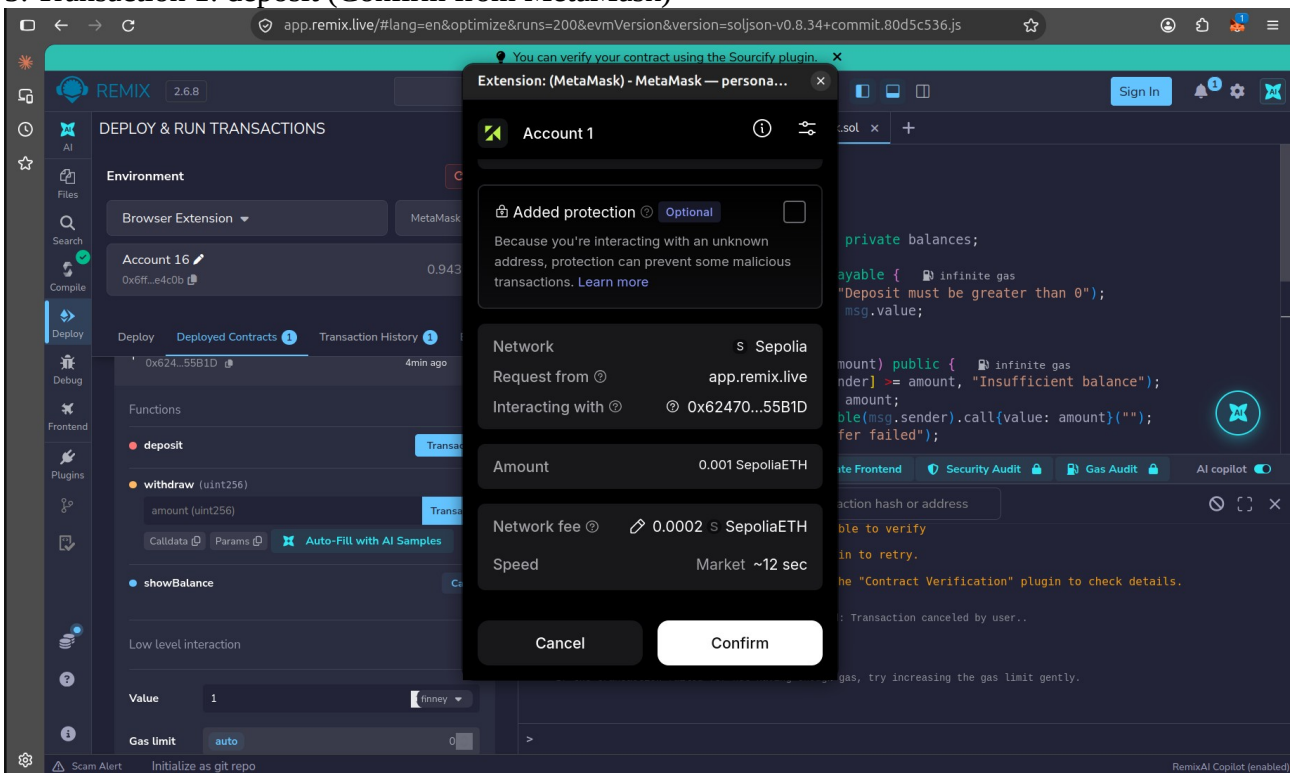
3. Deploy it



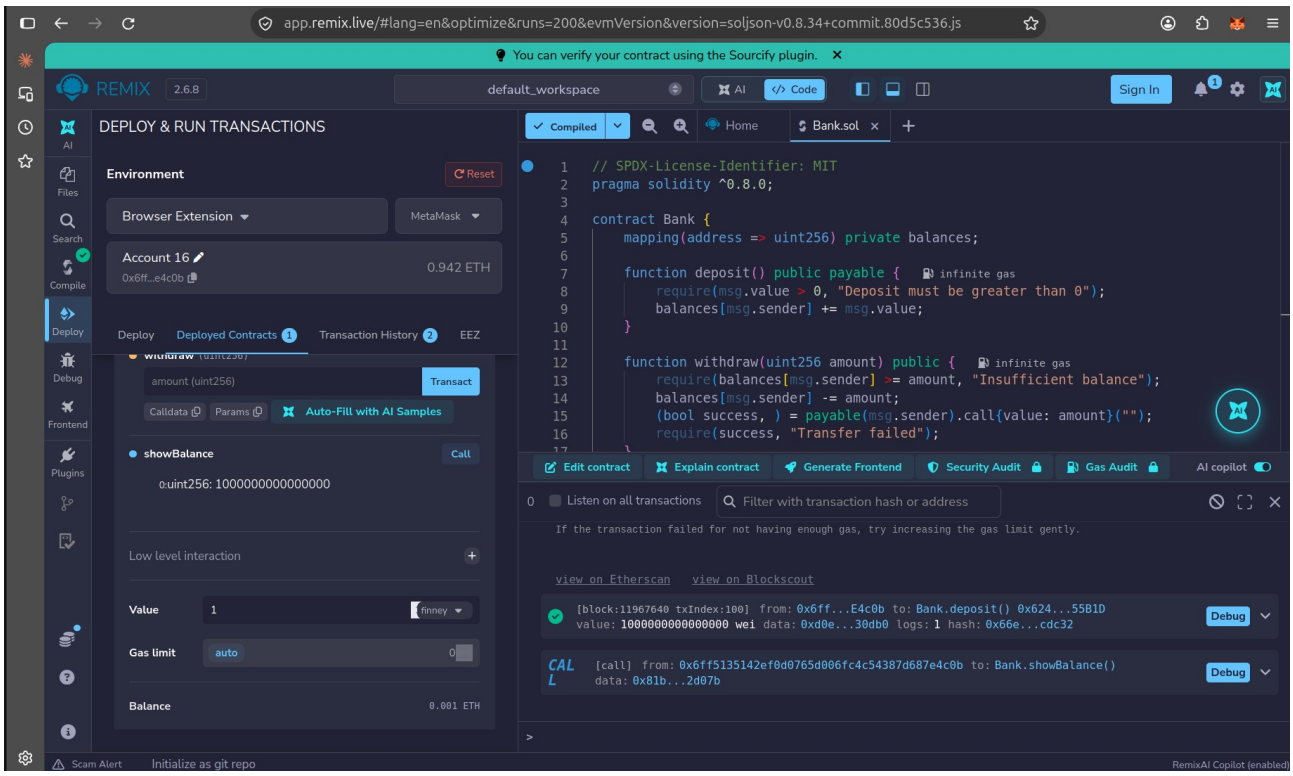
4. Connect MetaMask



5. Transaction 1: deposit (Confirm from MetaMask)



6. Transaction 2: show balance



7. Transaction 3: Withdraw

